Results

Tested on **6 fires the model never saw** (USA, Bolivia, Siberia, Greece).

In these fires 91% of cases are "fire spreads," so we compare against a simple baseline that **always guesses "spreads."**

Ranking risky cells (AUC-ROC, higher is better)

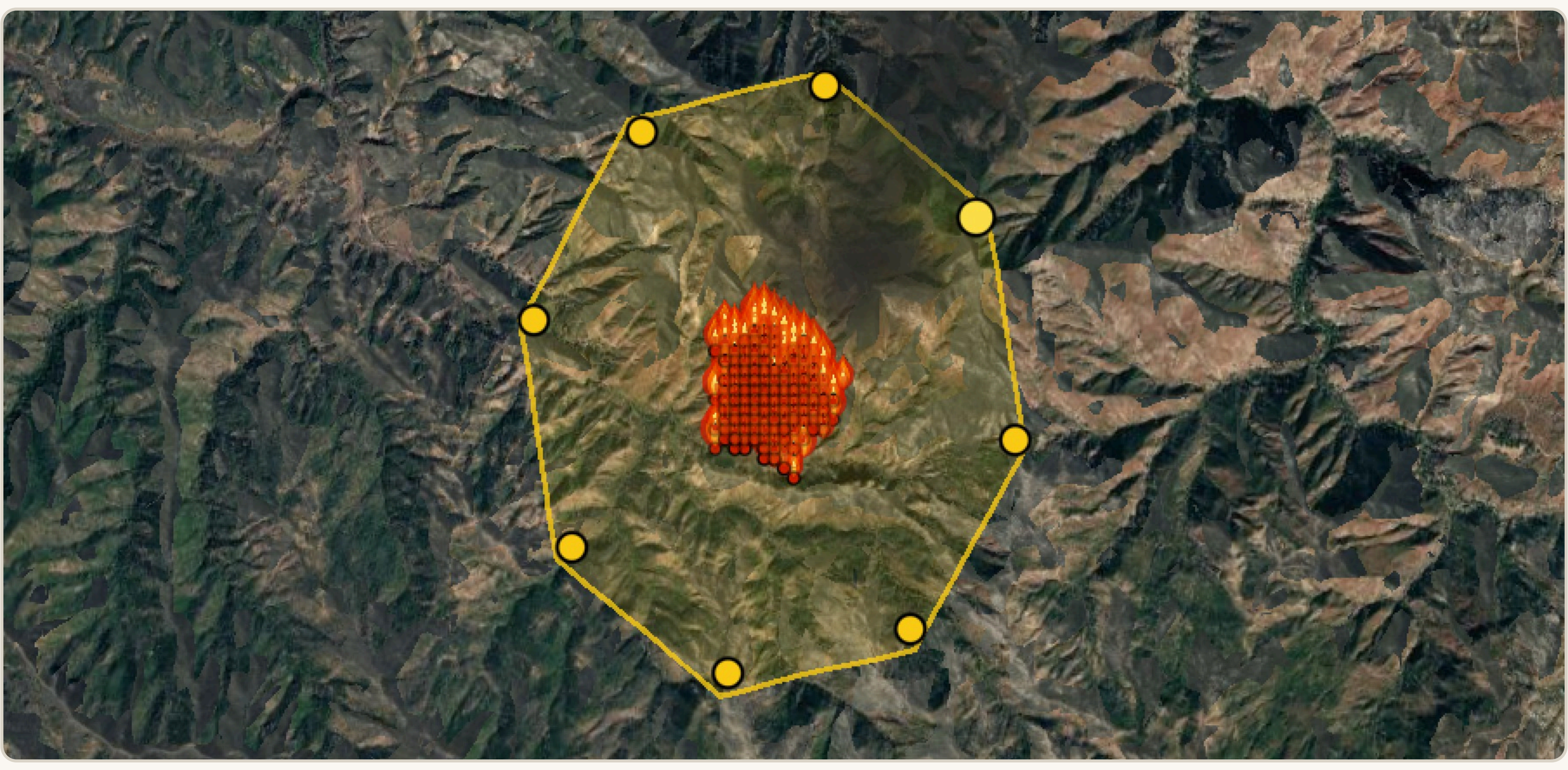
Our model	**0.75**
Baseline	**0.50**

Probability error (Brier score, lower is better)

Our model	**0.076**
Baseline	**0.083**

Takeaway: the model ranks which cells are at risk clearly better than chance, and its probabilities are more accurate than the baseline. It still leans toward "spreads," which we are working to improve.

Where to hold the line



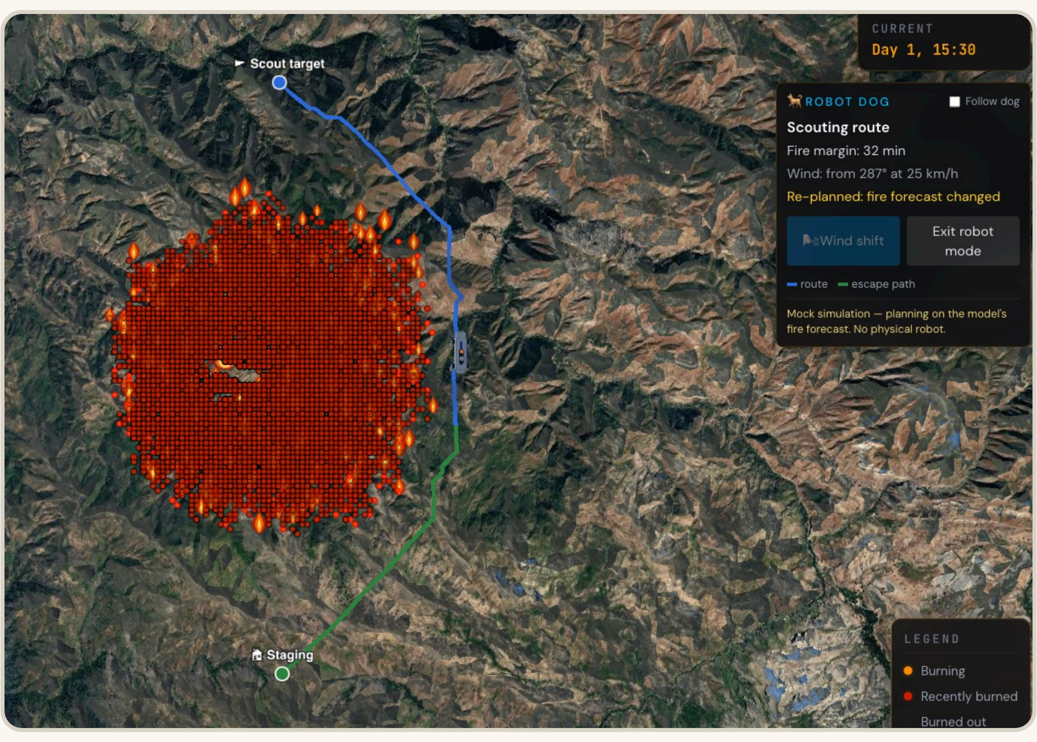
Pause the fire and draw a containment line. A **search algorithm** finds the **lowest-cost mix** of crews, engines, dozers and aircraft that keeps the fire inside it.

Example answer for this fire: **9 dozers + 1 helicopter crew + 1 air tanker.**

That line is exactly where the robot dog would scout (loop step 4).

Status & next

- ✔ ML fire forecast trained and tested
- ✔ 3D browser simulator on real terrain
- ✔ Containment planner (search)
- ✔ Robot-dog route planning on the forecast (mock simulation)
- → Run forecast + planner on the dog or its controller
- → Tests on real robot dogs



Scouting route

Fire margin: 32 min

Wind: from 287° at 25 km/h

Re-planned: fire forecast changed

Mock simulation — planning on the model's fire forecast. No physical robot.

Mock simulation: the robot dog re-plans after a wind shift. Its route (**blue**) bends around the forecast fire; an escape path (**green**) home is always kept. No physical robot.

Open-source code

Data & references

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections) · Open-Meteo weather archive · ESA WorldCover 10 m land cover · Cesium World Terrain · Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.
Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*